PEAD Strategy Research Summary

Signal-based study of post-earnings announcement drift using 2018-2026 U.S. earnings events

Sample	Strategy design	Primary conclusion
2018-2026; 651 high-score long signals in the full sample	Long-only PEAD screen on score = 4 earnings events; entry lag and holding period tested separately	The edge remained meaningful through 2024, but the payoff window shortened materially in 2025-2026

Executive Summary

- The core signal shows the classic PEAD shape in the full sample: performance is weak and noisy immediately after earnings, then improves steadily as the holding period extends. For the baseline long book, mean return rises from 0.17% over 1 day to 2.69% over 20 days, while win rate improves from 47.16% to 59.22%.
- Entry timing matters. Across the tested delays, entering on trading day +3 after the announcement produced the strongest 20-day outcome, with a 3.15% mean return and a 61.6% win rate. This is consistent with the idea that day 1 to day 2 price action still contains post-release noise, liquidity effects, and short-term overreaction.
- The signal is statistically meaningful rather than purely anecdotal. t-statistics remain above 2.0 across all tested configurations and strengthen as the holding horizon lengthens, with the entry_day = 3 curve reaching the strongest values.
- The annual profile is not uniform. The strategy worked best in 2019, 2020, 2023, and 2024, while 2025 marked a visible deterioration in longer-horizon returns. Early 2026 extends that weakness, although the 2026 sample is still small and should be interpreted cautiously.
- The most important research takeaway is not simply that PEAD worked, but that the market response appears to be accelerating. The evidence suggests a regime shift: what previously looked like a multi-week drift increasingly behaved like a multi-day adjustment in the most recent period.

Research Setup

This study evaluates a post-earnings announcement drift strategy built on high-conviction earnings signals. The portfolio is formed by selecting score = 4 events and taking long positions only. Performance is tracked across multiple holding horizons and entry delays in order to separate the effect of signal quality from execution timing.

The full sample spans 2018-2026. The baseline evaluation compares 1-day, 3-day, 5-day, 10-day, and 20-day holding periods. A second layer of analysis tests whether waiting to enter after the earnings release improves the quality of the trade. Finally, results are broken down year by year to evaluate stability, parameter sensitivity, and potential market structure change.

Full-sample performance

Hold	Trades	Mean return	Median return	Win rate

1 day	651	0.17%	-0.18%	47.16%
3 days	651	0.49%	0.10%	51.00%
5 days	649	0.73%	0.83%	54.85%
10 days	646	1.32%	1.32%	56.19%
20 days	645	2.69%	1.87%	59.22%

The progression is economically intuitive: the 1-day result is only marginally positive and the win rate remains below 50%, while the 10-day and 20-day horizons show a much cleaner payoff profile. That pattern is consistent with earnings information being incorporated with delay rather than fully priced on the announcement date.

Signal quality and statistical strength

The two figures below summarize the parameter sweep across entry delays and holding periods. Both the economic payoff and the statistical strength improve with longer holding periods, and the entry_day = 3 configuration consistently dominates the alternatives.

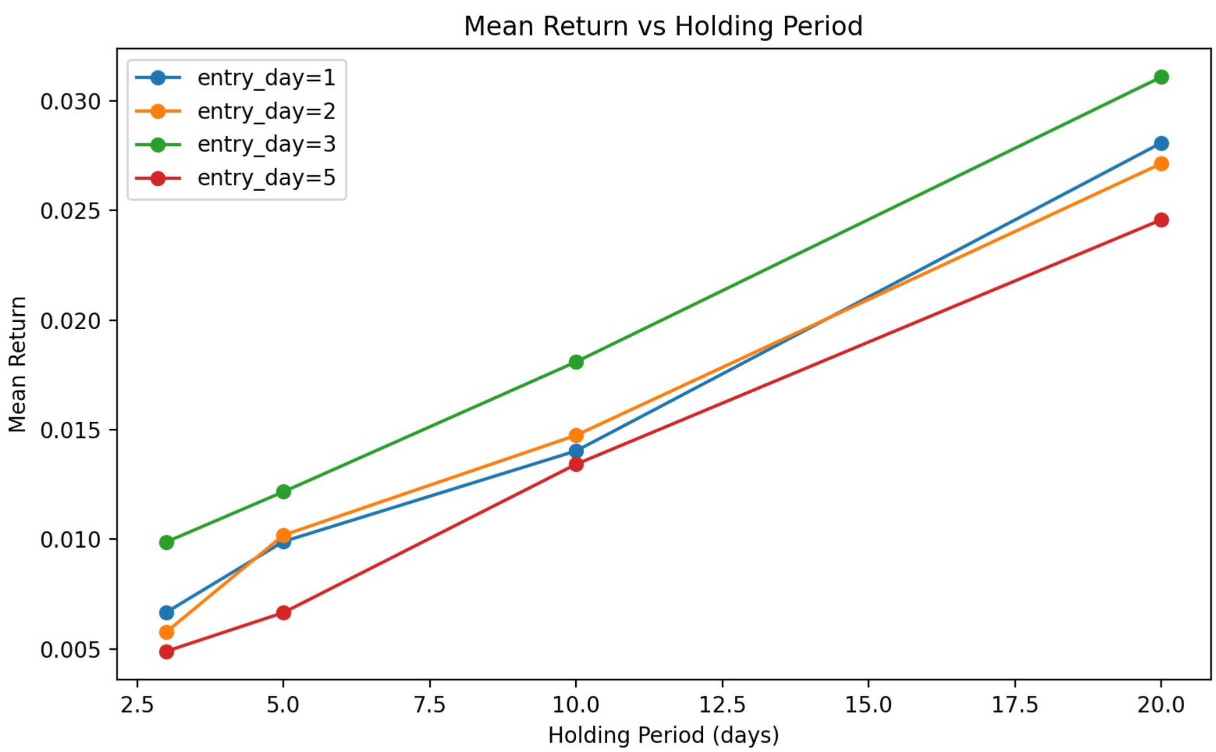

Figure 1. Mean return by holding period and entry delay. The entry_day = 3 configuration produces the strongest return profile at every medium-to-long horizon.

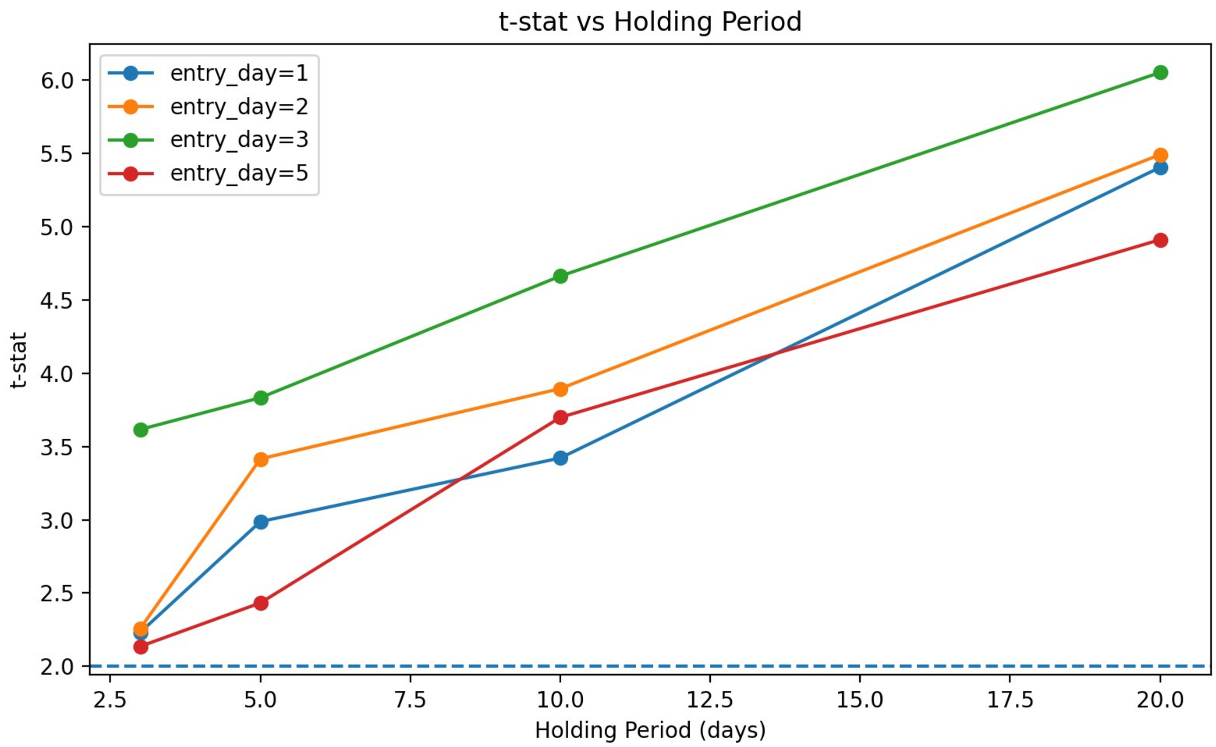

Figure 2. t-statistics by holding period and entry delay. Statistical strength remains above the 2.0 threshold across all tested combinations and peaks for entry_day = 3 at longer horizons.

A useful interpretation is that the signal does not simply produce one isolated good parameter. Instead, the strategy remains directionally sound over a range of horizons, with entry_day = 3 emerging as the cleanest execution rule. That kind of consistency matters because it suggests the result is less likely to be an artifact of one narrow backtest setting.

Entry Timing Analysis

For the 20-day holding horizon, the tested entry delays produce the following outcomes:

Entry delay	Holding period	Mean return	Win rate
1 day	20 days	2.82%	58.8%
2 days	20 days	2.71%	58.3%
3 days	20 days	3.15%	61.6%
5 days	20 days	2.51%	60.5%

The fact that entry_day = 3 outperforms entry_day = 1 and entry_day = 2 suggests that the first two sessions after earnings still contain substantial noise. Waiting a few sessions appears to preserve most of the drift while reducing the impact of short-term dislocation.

Year-by-Year Stability

The annual decomposition is the most important robustness check in this project. It shows that the strategy was not uniformly strong every year, but it was meaningfully effective for an extended window before weakening sharply in the most recent period.

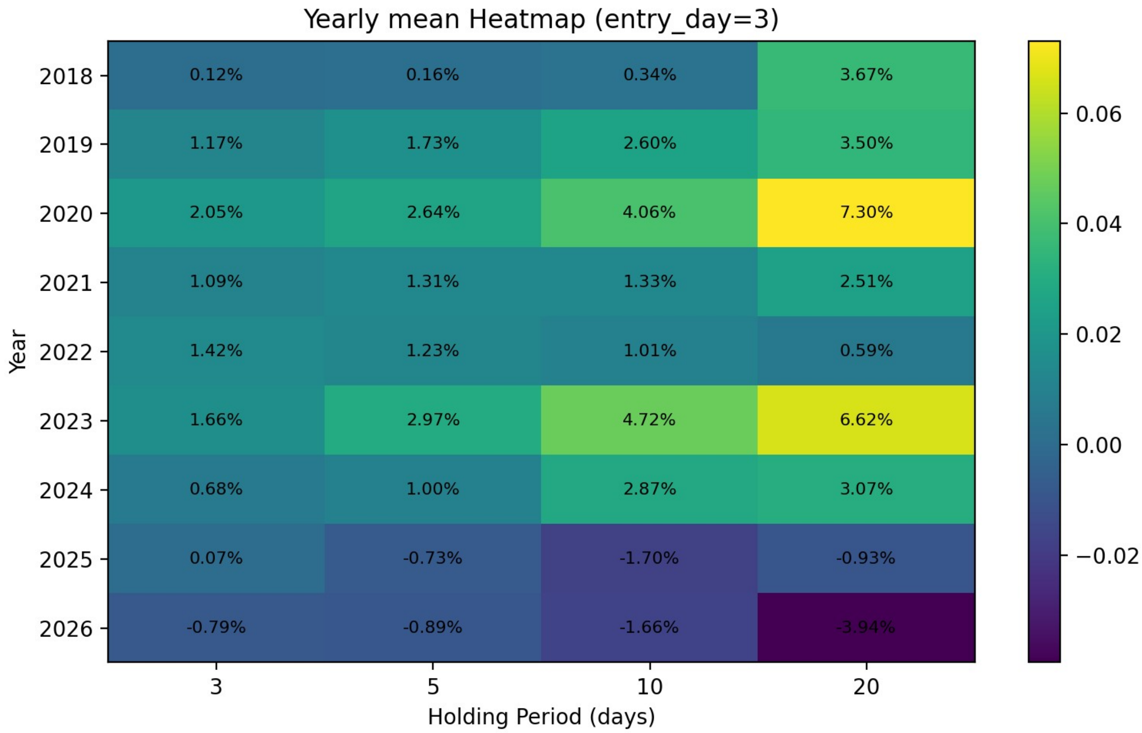

Figure 3. Yearly mean return heatmap for entry_day = 3. Earlier years show a clear increase in return as the holding period lengthens; 2025 and 2026 reverse that pattern, indicating that the long-horizon drift window has compressed.

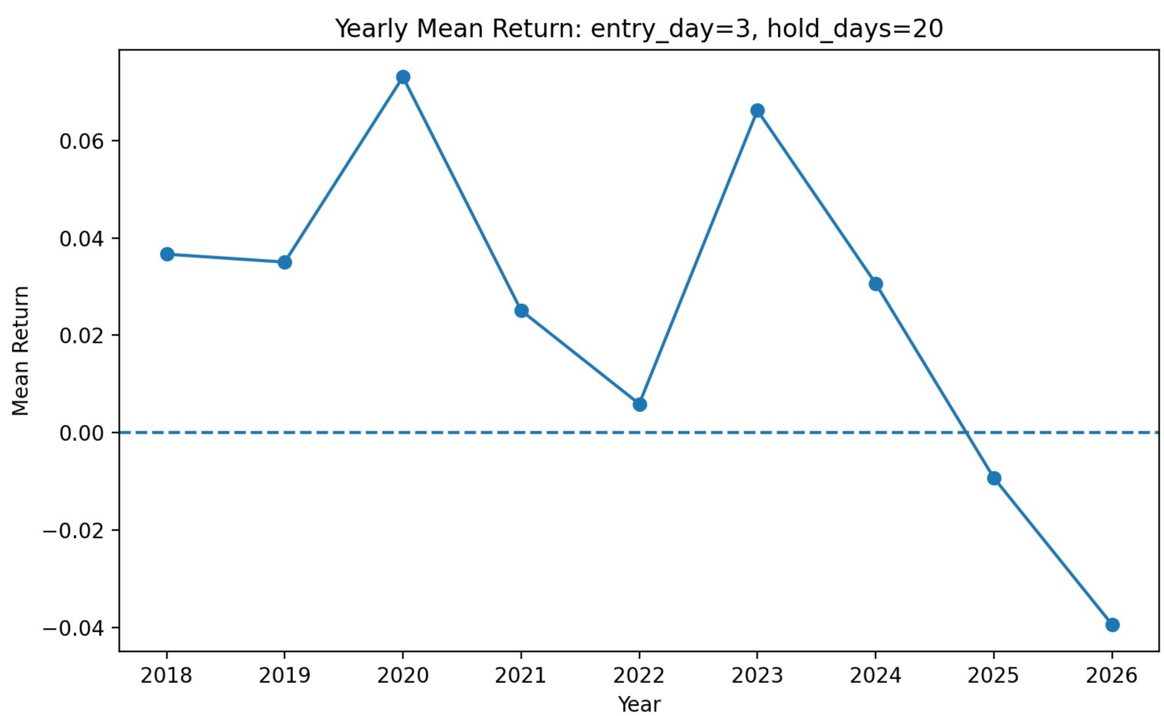

Figure 4. Yearly mean return for the historically best configuration (entry_day = 3, hold_days = 20). Returns remain positive through 2024, then turn negative in 2025 and deteriorate further in the early 2026 sample.

The annual results point to three distinct phases. First, 2018-2021 shows a workable but uneven drift profile. Second, 2023-2024 is the strongest stretch, with very attractive long-horizon returns. Third, 2025-2026 shows a clear loss of edge in the 10-day and 20-day windows, while shorter holds remain relatively more resilient. The natural interpretation is that the market is still responding to earnings, but it is doing so faster than before.

Selected 20-day annual results (baseline portfolio)

Year	Sample size	Mean return	Win rate
2018	69	2.82%	62.32%
2019	75	3.88%	65.33%
2020	81	4.38%	66.67%
2021	81	1.65%	56.79%
2022	80	1.06%	53.75%
2023	75	5.69%	62.67%
2024	87	4.00%	63.22%
2025	88	-0.16%	50.00%
2026	9	-9.17%	11.11%

Implementation and Transaction-Cost View

To translate raw signal performance into something closer to deployable PnL, the strategy was also simulated with an initial capital base of $2,000 and Interactive Brokers-style transaction costs of 0.7% per round trip. That exercise reinforces the same conclusion: the strongest years were 2022-2024, while the long-horizon version broke down in 2025.

Year	Hold	Signals	Ending capital	Annual return
2022	3d	80	$3,416	+70.8%
2022	5d	80	$4,260	+113.0%
2022	10d	80	$3,799	+90.0%
2022	20d	80	$2,008	+0.4%
2023	3d	75	$4,710	+135.5%
2023	5d	75	$5,890	+194.5%
2023	10d	75	$8,589	+329.4%
2023	20d	75	$20,483	+924.2%
2024	3d	87	$2,289	+14.4%
2024	5d	87	$2,489	+24.4%
2024	10d	87	$3,774	+88.7%
2024	20d	87	$4,433	+121.6%

2025	3d	88	$2,364	+18.2%
2025	5d	88	$2,261	+13.0%
2025	10d	88	$1,206	-39.7%
2025	20d	88	$1,036	-48.2%

The transaction-cost simulation is helpful because it prevents the analysis from relying only on abstract average returns. Even after costs, the 2023 and 2024 results remain compelling, while the 2025 deterioration is severe enough that the older 10-day and 20-day templates would have become economically unattractive.

Interpretation and Next Research Steps

The strongest version of the result is not that PEAD always works. The stronger claim, and the one better supported by the evidence, is that this earnings signal captured a real medium-horizon drift through most of the sample, but the timing of that drift changed materially in the latest regime.

That distinction matters. A strategy can lose money not because the underlying intuition disappears, but because the market learns to process the same information faster. In this project, 2025 appears to be the transition year: longer holds stop working, while shorter horizons remain comparatively more stable. The implication is that future work should focus on faster execution, narrower holding windows, and regime-aware parameter selection rather than treating the historical best setting as static.

From a research standpoint, the main contribution of this project is therefore twofold: first, it documents a statistically meaningful PEAD-style edge in a large multi-year sample; second, it identifies evidence that the edge is decaying in duration, which is a more decision-relevant finding than a simple in-sample performance summary.

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